

Design and Analysis of Algorithms

CSIT — 5th Semester — 2082 — Answer Sheet
Subject Code: CSC325

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. How do you define optimal solution? Does greedy algorithm always guarantee optimal solution? Given the string "SUPER DUPER CSIT", use a Greedy algorithm to build a Huffman tree.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is order statistics? Write and analyze the algorithm for randomized quick sort.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Distinguish between dynamic programming and memorization. Parenthesize the matrices $A(30 \times 1)$, $B(1 \times 40)$, $C(40 \times 10)$ and $D(10 \times 15)$, for computing matrix multiplication using dynamic programming.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Solve the recurrence relation $T(n) = 2T(n/2) + n$ using recursion tree method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Find the best and worst case for Bubble sort.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Using Extended Euclidean Algorithm, find the GCD of 12 and 16.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Find all possible subsets of the integers that sum to 21 in the array {5, 6, 10, 11, 15} using back tracking technique.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define class P and NP problem. Why do we need approximation algorithms? Justify.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. State the time and space complexity for sequential search. Write the rules for master theorem for finding asymptotic bounds.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Justify the worst case for binary search. Find the edit distance from the string "RELEVANT" to "ELEPHANT" using dynamic programming approach.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Distinguish between recursion and backtracking. Using Miller-Rabin primality test, check whether 53 is prime or not?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. How does 0/1 Knapsack problem differ with fractional one? Find the minimum vertex cover in the following graph.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.